

How to prevent

Office Ergonomics and Occupational Overuse Syndrome

Symptoms | Treatments

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What is - Office Ergonomics and Occupational Overuse Syndrome

Overview of Office Ergonomics

At the end of the workday, you uncurl yourself from your seat in front of your computer screen and you find your shoulder and back hurts, and there is a nagging pain in your hands. This could be the result of poor ergonomics and the way you work.



Ergonomics is the science of fitting the job to the worker. When there is a mismatch between the physical requirements of the job and the physical capacity of the worker, Occupational Overuse Syndrome (OOS) can result. Occupational Overuse Syndrome, also known as Repetitive Stress Injury (RSI), is a collective term for a range of conditions characterised by discomfort or persistent pain in muscles, tendons, and other soft tissues. Some common conditions are backaches, stiff necks and shoulders and carpal tunnel syndrome.

Ergonomics is a way to solve a host of physical problems brought about by over-exertion or repetitive stress.

Office Ergonomics and Occupational Overuse Syndrome - How to prevent

What can be done to prevent Occupational Overuse Syndrome?

Prevention is better than cure. Often OOS can be prevented by simple and inexpensive changes in your workplace. It is important to design the job, tasks and the work environment, including equipment and furniture, so that the factors that contribute to the risk of injuries can be avoided or minimised. It is also important that you are aware of the ergonomic principles, such as your working postures.

Workstation Design

The main components of an office workstation include the desk, chair and the equipment used to perform office task, often a computer. When designing a workstation, it is important that you consider task that you have to perform, and the type of equipment used. Flexibility is the key in workstation design.

Posture and Movement -- The way the workstation is designed affects the posture you adopt while working. Therefore, the workstation should be designed to place you in the best position to enable you to perform your work in comfort.

Chair -- An adjustable chair should be used for office work. It is necessary to adjust the chair to fit your body. You should also try to avoid sitting for long period of time. Some form of break from sitting every 30 minutes is beneficial.

Height -- adjust the chair height so that the feet are comfortable flat on the floor; the thighs are approximately horizontal and the lower leg approximately vertical. A footrest may be used if the feet are not touching the floor. Avoid wearing high heels. Tilt (if available) – set to horizontal initially, although this may be adjusted to tilt slightly forward to suit your comfort.

Height of Back Support – set the back support to rest at the curve of the lower back.

Forward/backward Position of Back Support – adjust the position of the back rest until a comfortable pressure is exerted on the low back area while seated in the usual working posture at the desk. The backrest position should not feel as though it pushes you out of the seat or that you have to lean back too far to reach it. There should be a two-finger clearance between the front of the chair and the back of the knee.

Armrests – armrests are usually not essential unless they are well out of the way of the desk. If you are using the arm rests, make sure that they do not prevent you from getting as close to the desk as you require. The height of the armrests should not be too high as to cause your shoulder to be elevated when the arms are rested. They should also not be too low where you have to tilt your body to rest your arms on them.

Desk -- Having adjusted the chair to suit your body size, adjust the height of the desk so the top surface is just below elbow height. The elbow height is determined by relaxing your shoulders and bending your elbows to about 90° and checking the elbow height against the desk height.

As most offices do not have height-adjustable desks, it may be necessary to adjust the height of the chair instead. If the desk is too high, raise the height of the chair so that your elbow height is just above the top surface of the desk. You may then require a footrest to rest your feet so that they are not left dangling above the floor.

Clearance under the desk -- Keep the space under the desk clear. Do not store bulky items as they will decrease or interfere with the space required for the legs. This may force the person to adopt a twisted or awkward posture of the spine.

Computer Monitor -- Once the desk and chair has been set, the monitor should be positioned. The monitor should be positioned so that the top of the screen is level with or slightly lower than your eye level. You may need a monitor stand, the CPU case, or something similar to achieve this.

The distance between the monitor and the eyes should be about one arm's length away from the user. The screen should be placed directly in front of the user in order to prevent unnecessary twisting of the body, which would place unnecessary strain on the back and neck. Avoid placing the screen opposite windows to avoid glares and reflections.

Key Board Angle – tilt the keyboard, using the feet at the back, to suit your level of comfort. The preferred setting is where the feet are lowered so the keyboard sits flat on the desk. When typing on the keyboard, your wrist should not be bent.

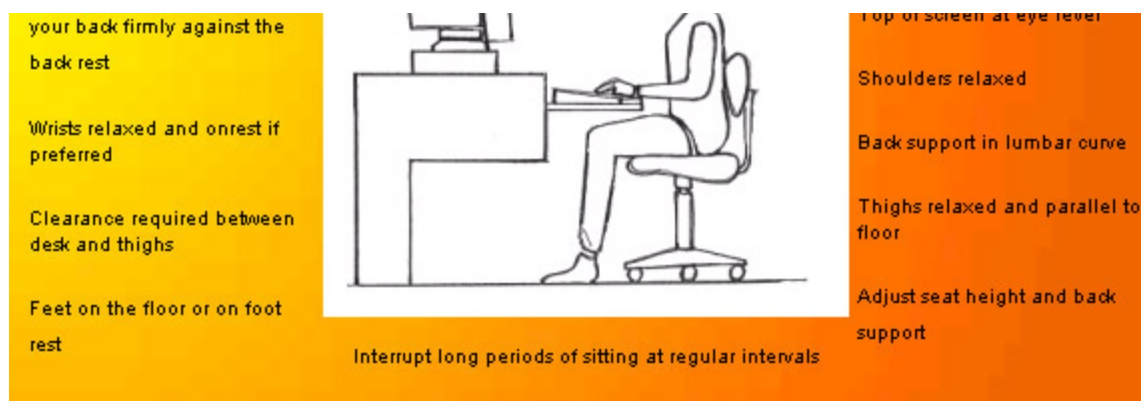
Mouse -- The mouse mat should be placed directly beside the end of the keyboard on your preferred side. Set the tracking speed of the mouse to a setting that suits you. Where possible, avoid holding onto the mouse when not in use.

Workstation Checklist

Sit squarely in the chair with



Feet supported on the floor



Stretching Exercises in the Workplace

Why Stretch?

Sitting for long periods can cause postural tension, muscle aches and even pain. Taking 5 to 10 minutes break relieves strain and tension and can revitalise your whole body.

This series of exercises have been designed to reduce areas of muscular tension in your body and help you to increase your overall flexibility. If you have a medical condition that could limit you from performing these movements, contact your treating practitioner before attempting these stretches.

How to stretch

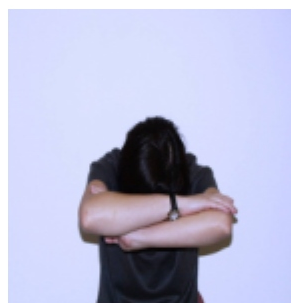
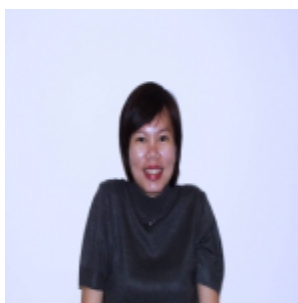
Stretching should be performed slowly bouncing or jerking. As you stretch, the tension should begin to ease. After this, you can gently begin to increase the stretch.

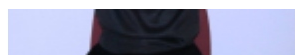
Remember to breathe normally as you begin your stretches.

If the stretches are painful, or if pain and tension persist on a daily basis, you may have muscle, joint or nerve problems and should consult a treating therapist.

- **Shoulder Roll**

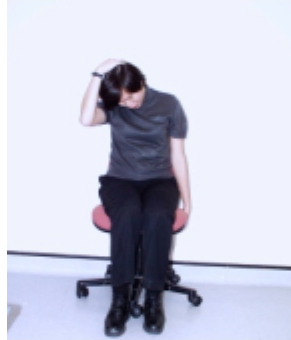
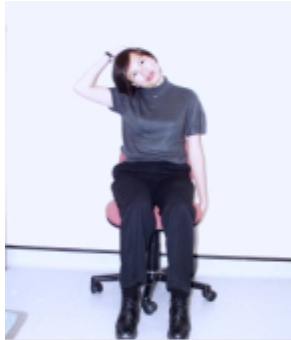
Shrug shoulders. Hold for 7-10 seconds. Gently move the shoulders back, down and forward. Raise the arms to shoulder height, crossing them at the elbow. Gently lower the head. Hold for 10 seconds.





- Chair Neck Stretch

Sit in chair. Hold on to the back legs with the right hand to steady right shoulder. Gently bring your left ear to your left shoulder until you feel the tension. Hold until the tension begins to diminish and gradually increase stretch. Hold for 10 seconds. Turn your head to the left and gently lower your head to look downwards (at your left shirt pocket). Hold. Increase stretch. Hold. Repeat for the other side.



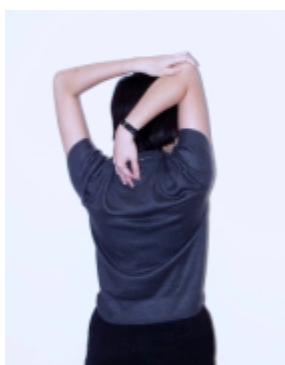
- Doorway Stretch

Stand in a doorway with your elbow flexed, holding on to the door frame at elbow height. Gently lean through the doorway until you can feel tension in the shoulder blade or chest region. Hold for 20 seconds.

Repeat exercise with hands at ear height, and again above head height.

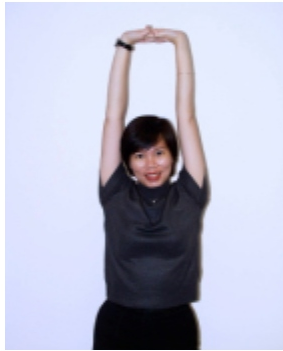


- Hold your right elbow with your left hand and gently pull on your right arm until you feel the stretching. Hold for 30 seconds. Do both sides.



- Interlace fingers and turn your palms upwards. Straighten your arms and stretch

Interlace fingers and turn your palms upward. Straighten your arms and stretch upwards. Pull both shoulder blades together and expand the chest. Hold for 15 seconds. Do this twice.



- Interlace fingers behind your back palm upwards. Turn your elbows inwards while you straighten both arms. Pull the shoulder blades together and expand the chest. Hold for 15 seconds. Do this twice



- Join hands above your head. Keeping your body facing straight ahead, bend towards your left. Hold for 15 seconds, then do other side. Do both sides twice.



Office Ergonomics and Occupational Overuse Syndrome - Causes and Risk Factors

Who are at risk of developing Occupational Overuse Syndrome (OOS)?

- People who must repeat the same motion throughout their workday
- People who must do their work in an awkward position
- People who must use a great deal of force to perform their jobs
- People who must repeatedly lift heavy objects
- Office workers who spend long hours working on the computers are also at risk of developing OOS if they do not observe good ergonomics

Contributed by

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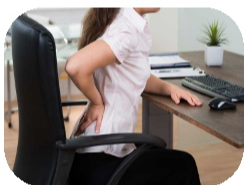
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